

Phase 5 Multi-Agent LangGraph — Detailed Test Case Report

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Executive Summary

Total Test Cases: **50** (spec-required 25 + additional 25)
Passed: **50** | Failed: **0** | Skipped: **0**
Overall Pass Rate: **100.0%**
Phase 5 Pass Threshold (≥18 of 25 spec cases): **CLEARED**

Suite-Level Summary

Class / Suite	Total	Passed	Skipped	Failed	Errors	Duration (s)
TestPhase5Spec25 (attached specification)	25	25	0	0	0	0.36
TestPhase5Additional25 (extra unit coverage)	25	25	0	0	0	4.36
TOTAL	50	50	0	0	0	4.72

Functional Category Breakdown

Category	Total	Passed	Failed/Skipped
Routing / Graph	9	9	0
State Schema	7	7	0
End-to-End	7	7	0
JSON Parsing Utility	6	6	0
Demand Forecaster Agent	5	5	0
Reorder Agent	5	5	0
Inventory Auditor Agent	5	5	0
Supplier Coordinator Agent	4	4	0
Agent Behavior	2	2	0

Section 1 — TestPhase5Spec25 (Attached Specification Suite)

Covers the 25 required Phase 5 test cases from the attached specification: state schema, agent behavior, routing logic, and end-to-end pipeline execution.

Agent Behavior (2 cases)

#	Test Case	Status	Time (s)	Details
1	test_12_agent_appends_messages	PASSED	0.007	-
2	test_24_api_failure_handled	PASSED	0.002	-

Demand Forecaster Agent (3 cases)

#	Test Case	Status	Time (s)	Details
1	test_05_demand_fetches_data	PASSED	0.002	-
2	test_06_demand_api_error	PASSED	0.002	-
3	test_07_forecast_fields	PASSED	0.003	-

End-to-End (6 cases)

#	Test Case	Status	Time (s)	Details
1	test_19_full_pipeline_executes	PASSED	0.003	-
2	test_20_audit_non_empty	PASSED	0.001	-
3	test_21_four_messages_in_trail	PASSED	0.003	-
4	test_22_terminal_status	PASSED	0.002	-
5	test_23_langsmith_optional	PASSED	0.001	-
6	test_25_urgent_reorder_status	PASSED	0.001	-

Inventory Auditor Agent (1 cases)

#	Test Case	Status	Time (s)	Details
1	test_11_auditor_generates	PASSED	0.001	-

Reorder Agent (2 cases)

#	Test Case	Status	Time (s)	Details
1	test_08_reorder_high_risk	PASSED	0.001	-
2	test_09_reorder_low_risk	PASSED	0.001	-

Routing / Graph (6 cases)

#	Test Case	Status	Time (s)	Details
1	test_13_normal_routes_reorder	PASSED	0.001	-
2	test_14_many_errors_route_audit	PASSED	0.001	-
3	test_15_error_status_routes_audit	PASSED	0.001	-
4	test_16_route_node_is_valid	PASSED	0.001	-
5	test_17_graph_has_four_nodes	PASSED	0.001	-
6	test_18_entry_node_is_demand	PASSED	0.001	-

State Schema (4 cases)

#	Test Case	Status	Time (s)	Details
1	test_01_state_fields	PASSED	0.002	-
2	test_02_initial_defaults	PASSED	0.001	-
3	test_03_no_mutation	PASSED	0.003	-
4	test_04_fields_persist	PASSED	0.001	-

Supplier Coordinator Agent (1 cases)

#	Test Case	Status	Time (s)	Details
1	test_10_supplier_quote	PASSED	0.002	-

Section 2 — TestPhase5Additional25 (Additional Unit Test Suite)

Covers 25 additional unit test cases created for deeper coverage: JSON parsing utility edge cases, state isolation guarantees, per-agent error handling, and router edge conditions.

Demand Forecaster Agent (2 cases)

#	Test Case	Status	Time (s)	Details
1	test_10_demand_keeps_existing_errors	PASSED	0.002	-
2	test_11_demand_keeps_existing_messages	PASSED	0.003	-

End-to-End (1 cases)

#	Test Case	Status	Time (s)	Details
1	test_25_pipeline_handles_model_failures	PASSED	0.001	-

Inventory Auditor Agent (4 cases)

#	Test Case	Status	Time (s)	Details
1	test_18_auditor_sets_complete	PASSED	0.002	-
2	test_19_auditor_fallback_report	PASSED	0.001	-
3	test_20_auditor_llm_error_surfaces_in_report	PASSED	0.001	-
4	test_21_auditor_appends_message	PASSED	0.003	-

JSON Parsing Utility (6 cases)

#	Test Case	Status	Time (s)	Details
1	test_01_safe_json_object	PASSED	0.001	-
2	test_02_safe_json_markdown	PASSED	0.001	-
3	test_03_safe_json_prose	PASSED	0.001	-
4	test_04_safe_json_empty	PASSED	0.001	-
5	test_05_safe_json_invalid	PASSED	0.001	-
6	test_06_safe_json_array	PASSED	0.001	-

Reorder Agent (3 cases)

#	Test Case	Status	Time (s)	Details
1	test_12_reorder_keeps_product	PASSED	0.002	-
2	test_13_reorder_false_preserves_analyzing	PASSED	0.001	-
3	test_14_reorder_true_sets_status	PASSED	0.002	-

Routing / Graph (3 cases)

#	Test Case	Status	Time (s)	Details
1	test_22_router_errors_below_threshold	PASSED	0.000	-
2	test_23_router_error_status_wins	PASSED	0.000	-
3	test_24_graph_is_invokable	PASSED	0.000	-

State Schema (3 cases)

#	Test Case	Status	Time (s)	Details
1	test_07_initial_state_isolated_lists	PASSED	0.001	-
2	test_08_initial_state_isolated_dicts	PASSED	0.001	-
3	test_09_product_id_preserved	PASSED	0.001	-

Supplier Coordinator Agent (3 cases)

#	Test Case	Status	Time (s)	Details
1	test_15_supplier_without_supplier_id_skips_fetch	PASSED	0.002	-
2	test_16_supplier_preserves_errors	PASSED	0.001	-
3	test_17_supplier_quote_has_fallback	PASSED	4.103	-